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Formation Mechanism of the Secondary Building Unit in a Chromium Terephthalate Metal–Organic Framework

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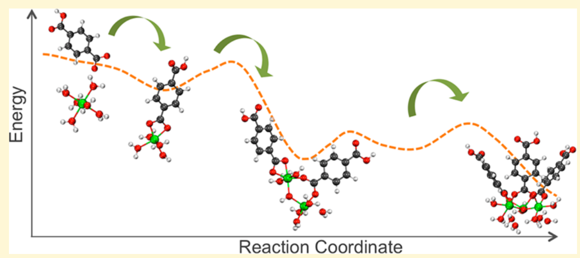
Supporting Information

ABSTRACT: A detailed mechanism, based on density functional theory calculations and simulation, is presented outlining the formation of the secondary building unit (SBU) of MIL-101, a chromium terephthalate metal organic framework (MOF). Formation of the metal core and of the SBU is key to MOF nucleation, the rate-limiting step in the synthesis process of many MOFs. A series of reactions that lead to the formation of the SBU of MIL-101 is proposed in this work. The highest barrier (~35 kcal/mol) involves the formation of a dimetal-linker intermediate and high to low spin transition as a third Cr-linker moiety joins to form a three metal-linker group joined by a central oxygen. The terephthalate linkers play an important, key mechanistic role with the carboxylates first joining chromium atoms prior to the formation of bridging oxygens. Subsequent to metal core formation, stepwise linker addition reactions generate different assembly pathways due to structural isomers that are limited by the removal of water molecules in the first chromium coordination shell. A simple kinetic model based on transition state theory gave a rate of SBU formation similar to a reported rate of MOF nucleation. The least energy path was identified with all linkers on the same face of the metal center added first. These first steps in developing a modeling framework for SBU formation will hopefully lay the groundwork for future comprehensive predictive models of the full MOF framework structure assembly and synthesis conditions required to support the self-assembly process.



■ INTRODUCTION

Metal–organic frameworks[1] (MOFs) are a class of materials well suited for gas storage[2] and separations.[3] If hydrothermal stability and cost challenges are met, they hold great promise for many applications including CO_2 capture,[4] heat transfer,[5] and building cooling.[6] MOFs are composed of metal or metal cluster centers joined by organic linkers through coordination bonds to form secondary building units (SBUs) that lead to the formation of a nanoporous structure. SBUs[7] have served as a way to understand MOF structure and topology, as they have in zeolites.[8]

Currently, MOF synthesis is a very uncertain art with most discovered through trial-and-error or automated[9−11] approaches rather than through a fundamental scientific understanding of the formation mechanisms. Consequently, the deliberate design of a MOF from basic components is highly uncommon. Known MOFs can be tailored toward desired structures and properties,[12] and new MOFs can be hypothesized based on empirical geometric assembly rules of known structures.[13] However, realizing MOF structures in practice is complicated because although thermodynamically stable structures may be predicted, they may not be kinetically accessible.

A recent critical overview[14] asserts the relevance that the coordination chemistry of basic building blocks, anions, and solvents have in MOF design, making crystal structure prediction and synthesis conditions greatly challenging. A thorough and precise knowledge of MOF synthesis pathways is currently inaccessible due to the complexity of metal-anion-solvent interactions and high number of competing simultaneous events that occur during MOF formation. The work in this paper outlines initial steps toward the ultimate goal of producing a first-principles predictive tool for MOF synthesis.

Few theoretical or experimental efforts have been focused toward a detailed stepwise understanding of MOF synthesis.[14−19] Two recent studies[20,21] found nucleation to be rate-limiting for copper-containing MOFs HKUST-1 and MOF-14 and aluminum-based MIL-53 and MIL-101 respectively with in situ X-ray data and a Gualtieri kinetic model,[22] which treats nucleation and growth distinctly. Another X-ray study[23] coupled kinetic data to separate Arrhenius models for nucleation and growth and found higher activation energies in the nucleation step for several MIL materials. This study revealed MIL formation rates proportional to the lability of the metal ions, suggesting that terephthalate linker deprotonation occurs faster than metal–linker complexation.

X-ray experiments have found discrete, fully formed SBUs during MIL-89 formation in solution prior to crystallization.[24]

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More recently, reactive intermediates prior to SBU formation and nucleation were found with in situ NMR techniques.[17] These studies highlight the importance of initial reactions and conditions for MOF synthesis, as SBU formation will affect MOF topology. Whether or not MOFs are formed by exclusively joining preformed SBUs,[15] SBU formation or the formation of inorganic metal center containing intermediates are critical for MOF synthesis.[17,25] This is likely the case in both the widely used "one-pot" synthesis approach where all components are mixed concurrently, as well as in proposed sequential or modular self-assembly synthesis of previously formed components.[16,26]

This work focuses on the initial reactions to construct an SBU because: (1) formed SBUs or metal centers are likely to be needed for nucleation, the rate-limiting process during MOF synthesis and (2) it is desirable to describe possible synthetic pathways of a complex process that is poorly understood involving multiple competing events affected by metal coordination chemistry, conformational isomers, and different spin-states. We have selected the MOF MIL-101[27] (Matériel Institut Lavoisier) as a model system for initial study due to availability of experimental nucleation kinetics data and interest in applications.[28,29] The SBU of MIL-101 is shown in Figure 1.

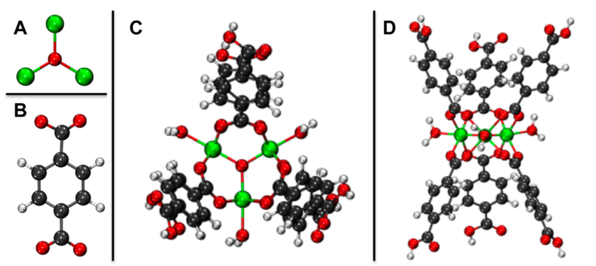

Figure 1. MIL-101, (A) the metal cluster, (B) terephthalate linkers, and the secondary building unit seen from above (C) and the side (D). Green is Cr(III), red is O, gray is C, and white is H.

The metal cluster is comprised of three octahedrally coordinated chromium(III) centers linked to a central bridging oxygen, with six carboxylate extension points. At the extension points, six terephthalate linkers (two on each side) join the three-metal cluster.

A series of chemical reactions, leading to the MIL-101 SBU, are presented starting from solvated chromium(III) ions $[Cr(H_2O)_6]^{3+}$ and partially deprotonated terephthalate linkers. Their associated mechanisms and energetics are revealed using density functional theory (DFT) calculations and ab initio molecular dynamics simulations (AIMD). This initial effort introduces basic reactions without the added complexity and ramifications that the anions from metal precursors present in the MIL-101 one-pot synthesis solution (NO_3^-, OH^-, or F^-) would bring. To our knowledge, this is the first computational attempt to uncover step-by-step reactions in the formation mechanisms of a full SBU, prior to nucleation, for MIL-101 or any MOF.

■ COMPUTATIONAL METHODS

In this work, **M** refers to the three chromium metal center $Cr_3(\mu_3\text{-}O)$, or cluster, and does not stand for metal or individual chromium, whereas **L** stands for the terephthalate linkers. These are depicted in Figure 1, along with the top and side view of the fully assembled MIL-101 SBU. All reactions leading to the SBU, starting from hexaaquachromium(III) ions $[Cr(H_2O)_6]^{3+}$ and terephthalate linkers, are displayed in Figure 2. Terephthalate, or benzene dicarboxylate linkers, were treated

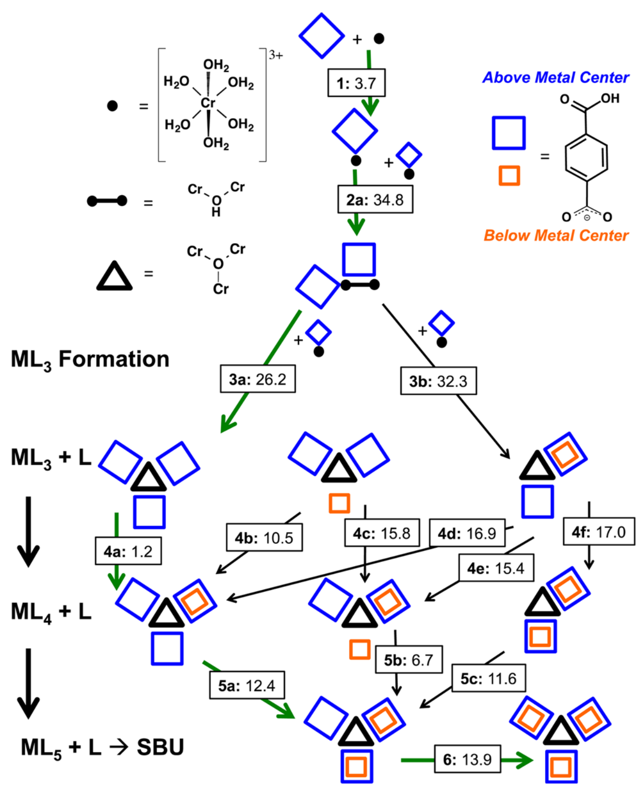

Figure 2. Reaction series that form the SBU of MIL-101. Reaction numbers in bold, energy barriers in kilocalories/mole. Not all water molecules shown explicitly for clarity. Least energy path in bold arrows. All Cr atoms are Cr(III).

with one end in its protonated form to cap the ends of the fully formed SBU. Visualization of the reactants and products was done with MacMolPlt[30] and VMD.[31] For the reaction paths shown in Figure 2, chromium atoms were octahedrally coordinated, and sites not bound to the central bridging oxygen or an **L** carboxylate oxygen included water molecules. This approach effectively includes first-solvation shell effects, which seem to have the biggest effect in the reactive barriers.

Spin-polarized density functional theory calculations and ab initio molecular dynamics (AIMD) were performed with the CP2K[32,33] code using the gradient-corrected Perdew, Burke, and Ernzerhof (PBE)[34] functional for exchange correlation. The norm conserving GTH pseudopotentials[35] model core electrons and a double-ζ quality basis set[36] for the valence electrons were used on all the atoms. A 340 Ry cutoff was used for the plane wave basis for the electrostatic energy.

Reactants were simulated with AIMD to explore possible reaction paths toward the products. The metadynamics[37] technique was used to accelerate AIMD sampling and cross high-energy barriers to visit other system configurations. For the reactions shown in Figure 2, metadynamics was used only to explore reaction mechanisms, not to fully reconstruct free-energy surfaces (FES), due to the complexity of collective variables and to the fact that fully converged FESs with multiple crossing events would be computationally too expensive. However, metadynamics was used to effectively simulate isomer transformations. Details on the metadynamics methods

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and the isomer transformation simulations appear in Sections 5 and 6, respectively, of the Supporting Information.

On the basis of the optimized reactant and product structures and the reaction paths observed in AIMD metadynamics simulations, reaction coordinate structures for all reactions in Figure 2 were constructed, followed by full geometry optimizations of all frames along the reaction coordinates to estimate the potential energy curve and barrier of each reaction. The reaction paths with the lowest energy barriers (Figure 2) were confirmed with climbing image nudged elastic band[38,39] (CI-NEB) calculations. CI-NEB minimizations used AIMD to optimize each reaction path, with a starting temperature of 10 K and an annealing factor of 0.95 and continued until maximum displacement (0.01 au, ~0.005 Å) and maximum force (0.004 au, ~4.5 kcal/mol·Å^{-1}) convergence criteria were met.

Local spin coupling during the SBU formation was further evaluated by considering different electronic spin states. To examine solvents effects beyond the explicit first-solvation shell, we also performed polarizable continuum model (PCM) calculations on select structures. Both sets of calculations were done with Gaussian09[40] using the PBE0 functional,[34,41] the 6-31G** basis set[42] for hydrogen, carbon, and oxygen, and the "Stuttgart RSC 1997" basis set and corresponding effective core potential for chromium, obtained from the EMSL basis set exchange[43] (https://bse.pnl.gov/bse/portal).

Finally, a simple kinetic model based on transition state theory (TST) was solved to estimate formation rates and activation energy of SBU formation. Coupled differential equations that represent the change in concentration for all species in Figure 2 over time were solved numerically, and are described in detail in Section 7 of the Supporting Information. For example, $\mathbf{ML}_5$ is produced by Reactions 5a, 5b, and 5c from different $\mathbf{ML}_4$ isomers and is consumed by Reaction 6 that produces the final SBU. Its concentration as a function of time is described by

$$\frac{\mathrm{d}[\mathbf{ML}_5]}{\mathrm{d}t} = k_{5a}[\mathbf{ML}_{4,i1}][\mathbf{L}] + k_{5b}[\mathbf{ML}_{4,i2}][\mathbf{L}] + k_{5c}[\mathbf{ML}_{4,i3}][\mathbf{L}] - k_6[\mathbf{ML}_5][\mathbf{L}]$$

where brackets denote concentration and k the rate constants.

Rate constants, at different temperatures, for each individual reaction were obtained based on a simple TST estimates using the energy barriers as determined in earlier steps. SBU formation rates at different temperatures were obtained by solving the coupled differential equations. Subsequently, the Arrhenius relation was used to obtain estimates of the activation energy and pre-exponential factor the reaction rates obtained from the kinetic model.

■ RESULTS AND DISCUSSION

Six reactive steps leading to a formed SBU of MIL-101 are proposed, shown in Figure 2. The energy barriers (E_B) shown for each reaction are calculated as the difference between reactants and the highest point on the potential surface along the reaction coordinate. Table S.1 in the Supporting Information summarizes binding energies, obtained as differences between the energies of the reactants and products in each reaction, as well as the energy barriers of each reaction. All products are thermodynamically favored over the reactant states. Binding energies calculated with a polarizable continuum model (Supporting Information Table S.2) are somewhat reduced compared to the gas phase; however, reactions still remain exothermic.

The first three reactions describe the formation of the $Cr_3(\mu_3\text{-}O)$ metal center with three terephthalate linkers $\mathbf{ML}_3$. In the subsequent reactions, linkers **L** are consecutively added on $\mathbf{ML}_3$ toward $\mathbf{ML}_6$, to form the full SBU. Because $\mathbf{ML}_3$ and $\mathbf{ML}_4$ have multiple isomers, Reactions 4 and 5 can proceed in different ways and all were considered. Figure 2 describes the isomer structures and the possible reactions. In Reaction 6, the final linker **L** is added onto $\mathbf{ML}_5$ to form the SBU. The reaction series in the path with the overall lowest energy barrier consists of Reactions 1, 2, 3a, 4a, 5a, and 6.

Transition metals are notorious for nondynamic correlation effects, arising from multiple nearly degenerate spin states. In our systems, the problem is compounded by mixed ligands (water and linkers), and broken-symmetry intermediate complexes as Cr(III) atoms are joined to form the SBU metal center. Although a single Cr(III) atom has 3 electrons resulting in a high-spin quartet ground state, subsequent addition of two more Cr(III) atoms produces a metal core with a total of six and nine d-electrons, respectively. Such spin populations in close vicinity are unstable. This in conjunction with structural deviations from high-symmetry molecular arrangement and Jahn–Teller induced instability ultimately favor low-spin states toward the late steps of the SBU formation. A very nice account of these effects on the spin states and magnetic properties of different metal clusters is given in the recent paper by Vogiatzis[44] and co-workers.

The two lowest electronic spin states of all reactive intermediate species were considered (Supporting Information Section 3). Prior to the formation of the trichromium metal center, the high spin quartet state for single chromium(III) species and the high spin triplet state for two-chromium(III) species are energetically favored over the doublet or singlet states respectively (Supporting Information Table S.3). Once the metal center has been formed with adjoining linkers ($\mathbf{ML}_3$), the energy difference between the spin states is negligible and eventually the system adopts the low-spin state. Because the quartet (or triplet) spin state is energetically favored prior to $\mathbf{ML}_3$ formation, the energy barriers obtained with CI-NEBs for Reactions 1 and 2 in the quartet and triplet spin states were chosen over those in the doublet and singlet spin states. Although both set of barriers (Supporting Information Table S.4) are in close agreement (within 6 kcal/mol), the barriers obtained with the quartet and triplet spin states are higher, and considered an upper bound for the formation of the SBU.

Rate Limiting Metal Center Formation Reactions. In Reaction 1 (Figure 3), a linker **L** substitutes two water molecules in the $[Cr(H_2O)_6]^{3+}$ coordination sphere and binds to the chromium forming a chelate compound $[Cr(H_2O)_4—\mathbf{L}]^{2+}$ where the chromium atom maintains its octahedral coordination. This ligand substitution reaction takes place with a dissociative mechanism: the water molecules leave the coordination sphere first, followed by ligand entry. Given the low energy barrier of this process (3.7 kcal/mol), an Eigen–Wilkins[45] mechanism for associative substitution appears unlikely as the formation of the $\mathbf{L}—[Cr(H_2O)_6]^{3+}$ encounter complex would be rate-limiting and not the leaving water molecules. This reaction proceeds rapidly, relative to the subsequent reaction, suggesting that the $[Cr(H_2O)_4—\mathbf{L}]^{2+}$ species could be detectable, with a lifetime on the order of 10^3 seconds as predicted by the kinetic model (Supporting Information Section 7). Quick $[Cr(H_2O)_4—\mathbf{L}]^{2+}$ formation

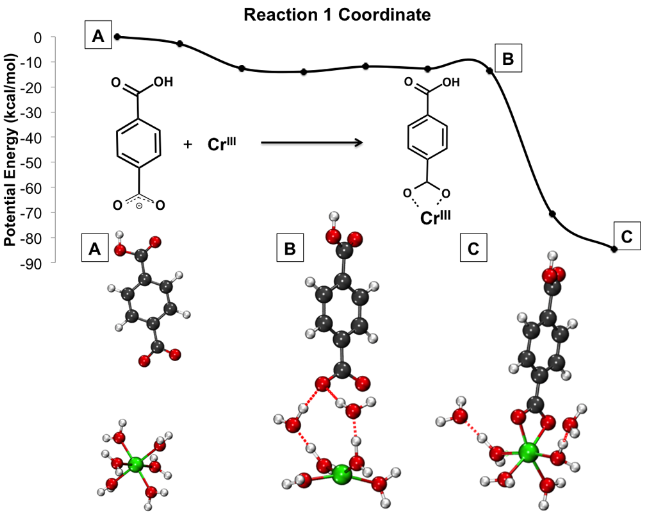

Figure 3. Reaction 1 coordinates and energy profile. Not all water molecules shown explicitly for clarity. Green is Cr(III), red is O, gray is C, and white is H. All Cr atoms are Cr(III).

also prevents chromium olation reactions that would obstruct MOF formation.

It is known that hexaaquachromium(III) ions are kinetically inert, with slow water exchange rates on the order of 10^{-3} s^{-1}.[46] Also, water exchange rates in MIL materials with different metals were found slowest with chromium, correlating with rates of chromium−linker formation.[23] Although this may slow water−linker substitution reactions, it is the chromium−linker strength that makes MIL-101 a relatively stable material.[29]

In the Reaction 2 (Figure 4), two $[Cr(H_2O)_4—\mathbf{L}]^{2+}$ (the product of Reaction 1) are combined to make compound

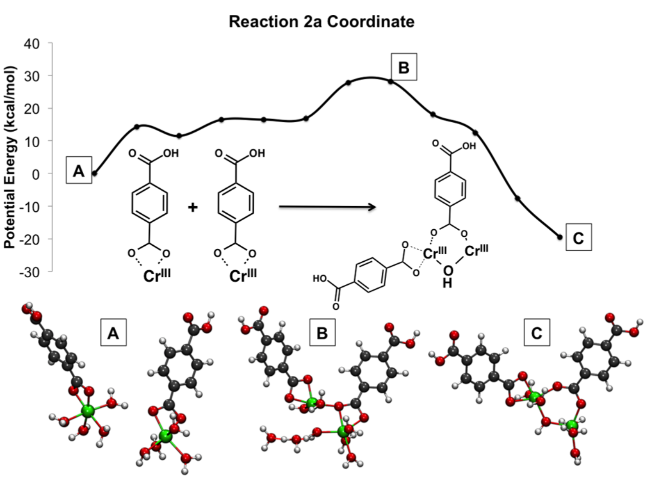

Figure 4. Reaction 2 coordinates and energy profile. Not all water molecules shown explicitly for clarity. Green is Cr(III), red is O, gray is C, and white is H. All Cr atoms are Cr(III).

$[Cr(H_2O)_4-(\mu-OH)-Cr(H_2O)_4—\mathbf{L}_2]^{3+}$ where a two-chromium hydroxo-bridged species accommodates one chelate and one bidentate ligand **L**. Hydroxo-bridged chromium species are known to form in mild acidic solutions,[46] and acidic solution conditions favor the formation of MIL-101.[27] In this reaction, first the linker in one $[Cr(H_2O)_4—\mathbf{L}]^{2+}$ binds with chromium in the other $[Cr(H_2O)_4—\mathbf{L}]^{2+}$ making a bis-monodentate intermediate, followed by the formation of the bridging hydroxide between the two chromium atoms. Although products are thermodynamically favored over reactants, this reaction has a relatively high energy barrier of 34.8 kcal/mol, the highest barrier in the lowest energy path in SBU formation.

The addition occurs in a dissociative type mechanism with a water molecule exiting the first-coordination shell of $[Cr(H_2O)_4—\mathbf{L}]^{2+}$. This is followed by the formation of the bond between the chromium in one $[Cr(H_2O)_4—\mathbf{L}]^{2+}$ and a **L** carboxylate oxygen from the other $[Cr(H_2O)_4—\mathbf{L}]^{2+}$. The high-energy barrier of this reaction results from this step. As one linker moves from a chelate conformation coordinated to a single chromium to a bis-monodentate conformation involving two chromiums, an additional water enters the chromium coordination shell in $[Cr(H_2O)_4—\mathbf{L}]^{2+}$ to keep octahedral geometry. Then, a water molecule acts as a base taking hydrogen from a water molecule bound to one chromium atom, making an OH group that quickly forms a bond with the other chromium, forming the bridging hydroxo-group and final product with one chelate **L** and a bidentate **L** joining the two chromiums. While a dehydration-type reaction where two hydroxide anions form the bridging oxygen and release water would be plausible, a baseline mechanism is presented here; added complexities and ramifications due to anions will be considered later.

Alternative mechanisms result in different isomers of the $[Cr(H_2O)_4-(\mu-OH)-Cr(H_2O)_4—\mathbf{L}_2]^{3+}$ product. Details can be found in Section 4 of the Supporting Information. In Reaction 2b (Supporting Information Figures S.1 and S.2), the bridging hydroxide first joins the two chromium atoms, leaving both **L**s in chelate coordination to each chromium atom. In Reaction 2c (Supporting Information Figures S.1 and S.2), both linkers join the chromium atoms prior to the formation of the bridging hydroxide ion resulting in a product with two bidentate **L**s. The resulting barriers for Reactions 2b and 2c are ∼15 and ∼35 kcal/mol higher than that of Reaction 2, respectively. The step where the bridging hydroxo group is formed causes the higher energy barrier in both alternative mechanisms. The presence of hydroxide ions is likely to lower the barrier of these steps, as the basic character of hydroxides promotes chromium-bound water activation for bridging hydroxo- or oxo- formation. Hydroxo-bridged dichromium species are converted to oxo-bridged species in the presence of hydroxide groups.[46]

Transformations between the different structural isomers of the $[Cr(H_2O)_4-(\mu-OH)-Cr(H_2O)_4—\mathbf{L}_2]^{3+}$ species were also simulated with the accelerated AIMD method metadynamics. Collective variables were based on coordination numbers between Cr atoms and O atoms of the incoming ligand. Simulation details and parameters are given in Sections 5 and 6.1 of the Supporting Information. Figure 5 shows the free energy profile of the conformations of one linker in the $[Cr(H_2O)_4-(\mu-OH)-Cr(H_2O)_4—\mathbf{L}_2]^{3+}$ species. The bidentate conformation, where the linker is coordinated to both chromium atoms, is preferred over the chelate conformation. Further metadynamics simulations also predict that linkers in the bidentate conformation are thermodynamically favorable; see Supporting Information Figures S.3, S.4.

Reaction 3a (Figure 6) describes how the products of Reactions 1 and 2, $[Cr(H_2O)_4—\mathbf{L}]^{2+}$ and $[Cr(H_2O)_4-(\mu-OH)-Cr(H_2O)_4—\mathbf{L}_2]^{3+}$, combine to form $[Cr_3(H_2O)_N-(\mu_3-O)—\mathbf{L}_3]^{5+}$ or $\mathbf{ML}_3$. Isomers 1 and 2 of $\mathbf{ML}_3$ with all sides of the trichromium metal center occupied are described by Reaction 3a. Similar to Reaction 2, the reactants are first joined by linkers followed by the formation of the bridging oxygen

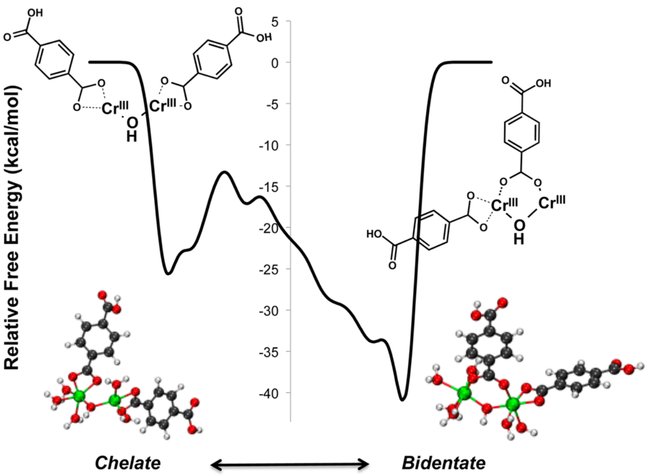

Figure 5. Free energy profile of the conformations one linker can take in the two-chromium intermediate. Not all water molecules shown explicitly for clarity. Green is Cr(III), red is O, gray is C, and white is H. All Cr atoms are Cr(III).

(μ_3-O). First the linker in $[Cr(H_2O)_4—\mathbf{L}]^{2+}$ moves from a chelate conformation to a bis-monodentate joining the two reactants in a dissociative-like mechanism where one chromium in $[Cr(H_2O)_4-(\mu-OH)-Cr(H_2O)_4—\mathbf{L}_2]^{3+}$ loses a water molecule and opens a coordination site for the carboxylate oxygen in $[Cr(H_2O)_4—\mathbf{L}]^{2+}$. Then, in the energy barrier defining step, the chelate **L** in $[Cr(H_2O)_4-(\mu-OH)-Cr(H_2O)_4—\mathbf{L}_2]^{3+}$ forms a bond with the chromium originally in the $[Cr(H_2O)_4—\mathbf{L}]^{2+}$ species, changing to a bis-monodentate coordination following also a dissociative-like mechanism. Finally, a water molecule takes the hydrogen from the bridging hydroxide, and the activated oxygen forms a bond with the remaining chromium making the metal center $Cr_3(\mu_3$-O) with three bidentate linkers at the carboxylate extension points of **M**.

Different Reaction 3 mechanisms will result in different $\mathbf{ML}_3$ isomer products. Further discussion on the different mechanisms and isomer transformations appears in Sections 4 and 6.2 of the Supporting Information respectively. Isomers 1 and 2 of $\mathbf{ML}_3$ are favorably produced directly by Reaction 3a.

Multiple Competing Linker Addition Reactions. $\mathbf{ML}_3$ has three possible structural isomers, and addition of another **L** to form $\mathbf{ML}_4$ can proceed in six different ways, labeled as Reactions 4a, 4b, 4c, 4d, 4e, and 4f in Figure 2. Similarly, in Reactions 5a, 5b, and 5c, **L** is added to $\mathbf{ML}_4$ to make $\mathbf{ML}_5$. In Reaction 6, a final **L** is added to form $\mathbf{ML}_6$.

All linker addition reactions proceed by the same dissociative-type mechanism, shown in Figure 7 for Reaction 6. First, as **L** approaches $\mathbf{ML}_\mathbf{N}$, two coordination waters move to the second solvation shell, initially forming hydrogen bonds to the first shell waters, and to the approaching carboxylate ligand. As **L** advances toward **M** the hydrogen bond network remains intact, until **L** is bound to the metal center **M**. The mechanism is dissociative as the water molecules first exit to allow linker entry, and the coordination of chromium atoms is reduced at the transition state. This behavior bears similarity to the Eigen−Wilkins mechanism for single metal octahedra ligand substitution in that the rate-limiting step will arises from the loss of the leaving ligand following the formation of an encounter complex.

The energy barrier of linker addition reactions result from displacing the water molecules coordinated to **M** as **L** approaches because Cr(III) ions have a large ligand stabilization energy, and water−linker exchange reactions disrupt ligand arrangement around chromium atoms in **M**. Although all linker addition reactions follow the same mechanism, differences in energy barriers among reactions arise not only from the different structural rearrangements of H_2O and ligands already bound to **M** but also from the change in electronic spin state. As the system crosses into the low spin state after the addition

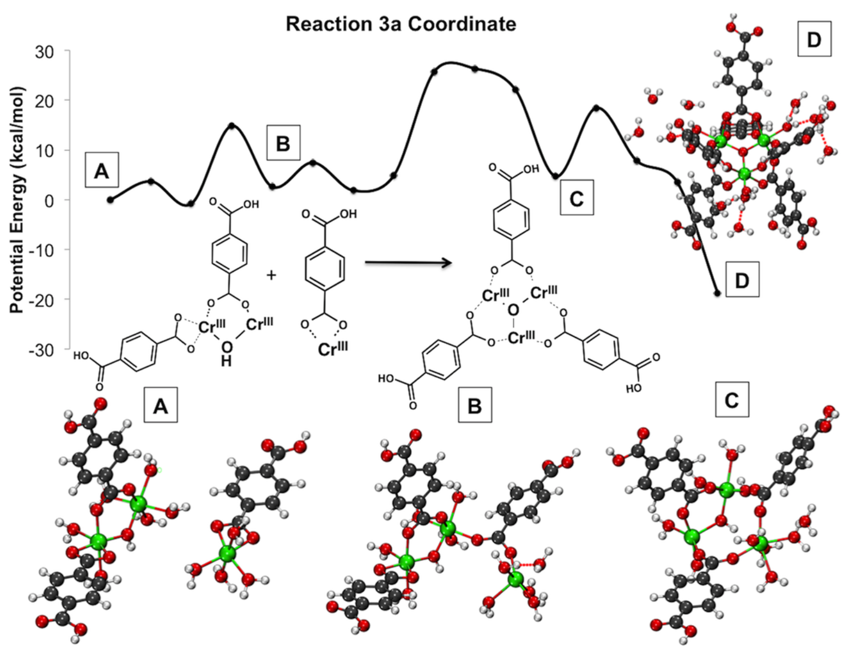

Figure 6. Reaction 3 coordinates and energy profile. In (D) the central bridging oxygen joining the three chromiums is formed, whereas in (C) an OH group joins two chromium atoms. Not all water molecules shown explicitly for clarity. Green is Cr(III), red is O, gray is C, and white is H. All Cr atoms are Cr(III).

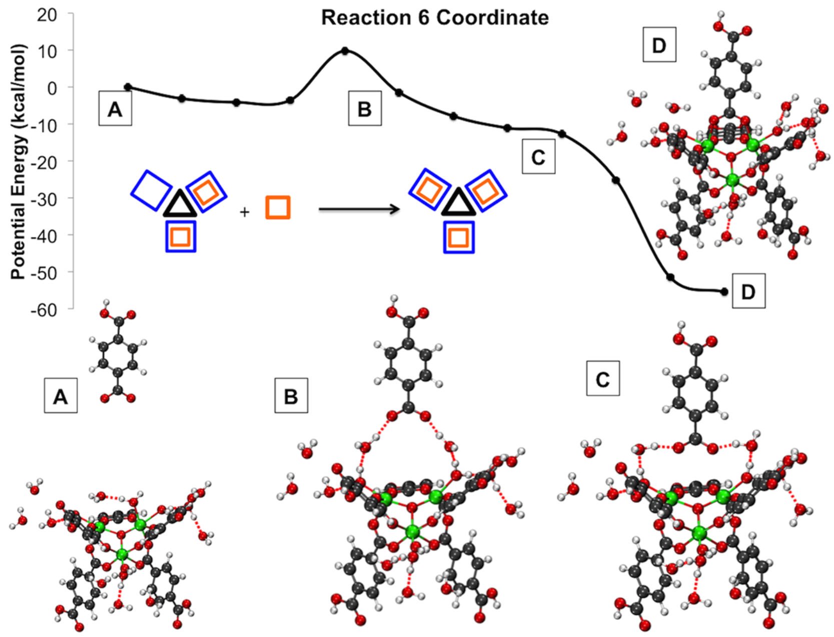

Figure 7. Energy profile of a linker addition reaction. Not all water molecules shown explicitly for clarity. Green is Cr(III), red is O, gray is C, and white is H. All Cr atoms are Cr(III).

of the third organic linker, the possible high and low spin states become isoenergetic.

Structural isomers of intermediates bring forth competing reactions toward the SBU. The reaction coordinates and energy barriers for the reactions in the lowest energy path were confirmed with CI-NEB calculations. In linker addition reactions, the CI-NEB energy barriers are very similar to the barriers found in optimized geometry reaction coordinates. Considering computational costs, only the reactions in the lowest energy path were confirmed with CI-NEBs, but the agreement between barriers in this path gives certainty to the barriers in other paths of the linker addition reactions estimated by simple linear path optimizations.

The path with the overall lowest energy barrier follows Reactions 1, 2, 3a, 4a, 5a, and 6 (bold arrows in Figure 2), and the path with the highest energy barrier Reactions 1, 2, 3b, 4f, 5c, and 6. In the lowest-energy path, first all linkers in the same side of the metal center plane are added followed by the linkers on the opposite side. In the highest-energy path, linkers are added on each metal-center edge, above and below the metal-center plane. This indicates high stereoselectivity that can lead to specific ligand substitutions that can be tailored to possibly tune the SBU properties.

Reaction 4a plays a significant role in determining the lowest energy path toward the formed SBU because its reaction energy barrier is considerably lower than other $\mathbf{ML_3}$ to $\mathbf{ML_4}$ reactions. In the $\mathbf{ML_3}$ isomer 1, there are bound linkers facing the same direction in all three sides of the metal-center plane. This reduces the number of coordination sites that water molecules can occupy on the opposite side while maintaining pseudo-octahedral coordination around the chromium centers. Overall, steric hindrance drives the removal of water as **L** approaches $\mathbf{ML_3}$ isomer 1 in Reaction 4a, resulting in a very small energy barrier for the reaction.

To determine how readily linkers can transition from above to below (or vice versa) the metal center plane, metadynamics AIMD simulations were done to simulate transformation between $\mathbf{ML_3}$ isomers 1 and 2 (Section 6.3 of the Supporting Information, Figures S.8 and S.9). From the free energy profile, the estimated barrier of 25 kcal/mol was more than twice the barrier of subsequent linker addition reactions, implying that this process is not likely, at least based on the thermodynamic criteria.

SBU Formation Rate and MOF Nucleation Rate. A recent study[23] fitted experimental nucleation rates to the Arrhenius equation and estimated the activation energy for MIL-53 nucleation (a chromium−terephthalate MOF) at 41.59 kcal/mol (174 kJ/mol) and a pre-exponential factor on the order of 10^{15} s^{-1} ($\sim10^{17}$ min^{-1}) from an experimental nucleation rate on the order of 10^{-5} s^{-1} (10^{-3} min^{-1}). Another study[47] from the same group compared MIL-53 and MIL-101 formation and concluded that both are produced at similar rates, though the specific rates were not calculated.

Using the kinetic model described in the Computational Methods section (with details in Section 7 of the Supporting Information), an SBU formation rate was predicted on the order 10^{-6} s^{-1}, 1 order of magnitude smaller than the nucleation rate of MIL-53 at the same temperature. Using rates from the kinetic model at various temperatures and the Arrhenius equation, an activation energy of 38.90 kcal/mol was estimated, slightly less than the observed experimental 41.59 kcal/mol barrier for MIL-53 nucleation. However, the Arrhenius relation, using rates obtained from the model, predict a pre-exponential factor in the order of 10^{12} s^{-1} for MIL-101, whereas the observed pre-exponential factor of MIL-53 nucleation is in the order of 10^{15} s^{-1}.

Our calculated energy barriers and kinetic model estimate SBU formation rates that qualitatively agree with experimental MOF nucleation rates. This may result in a systematic computational approach to reliably predict MOF nucleation rates from SBU structures.

Agreement with in Situ Experiments of MIL Materials. Recently, Férey and co-workers[17] and Goesten and co-workers,[19] performed in situ NMR studies on the formation of several MIL materials with a trialuminum core. Our proposed reactive series (Figure 2) for the trichromium SBU of MIL-101 is in qualitative agreement with their experimental findings.

Supporting that single metal-linker complexation is an important initial step, Goesten and co-workers[19] made two relevant observations: (1) there is an inverse correlation between Al-solvent coordination and linker concentration and (2) solvated Al^{3+} ions were consumed prior to the crystalline phase and during initial temperature ramping. Further, Férey and co-workers[17] actually detected $[Al(H_2O)_6]^{3+}$ and $[Al(H_2O)_N(H_2\mathbf{btc})]^{2+}$ species, where **btc** is the benzene tricarboxylate linker. Similar to our proposition that the $[Cr(H_2O)_4—\mathbf{L}]^{2+}$ species (produced by Reaction 1) is needed for synthesis, they also find that 1:1 metal−ligand species are initially present and necessary for MOF formation.

Férey and co-workers[17] also detected dimetal complexes, $Al_2-(\mu-OH)-(\mathbf{btc})_2$, which correspond to the two-chromium species $[Cr(H_2O)_4-(\mu-OH)-Cr(H_2O)_4—\mathbf{L_2}]^{3+}$ produced by Reaction 2. They refer to these species as prenucleation building units (PNBUs). After the aluminum PNBUs reached a critical concentration, zero-charged complexes (corresponding to the $\mathbf{ML_3}$ species in our work) are formed, and nucleation quickly follows. This supports our kinetics conclusion that SBU formation rates, which are rate-limited by metal-center formation, strongly correlate to MOF nucleation rates.

Their work also aligns with our mechanistic proposal that carboxylate groups join metal atoms first, before they form the metal center. Férey and co-workers[17] found the trimetal center with bridging oxygen formed throughout the hydrothermal reaction of an iron MIL material. However, unlike MIL-101 and the aluminum MIL materials they also studied, the metals (iron) were added with acetate anions prior to linker addition. The carboxylate groups in the acetate anions could play the role that the carboxylate groups in terephthalate linkers for MIL-101 during the initial reactions to join metal ions and form the bridging oxygen.

In this work, we propose a mechanism for the formation of an SBU, but a MOF nucleation mechanism remains unclear. It is to be determined whether nucleation involves joining PNBUs, neutral species (corresponding to our $\mathbf{ML_3}$), full SBUs, or combinations thereof. However, resolving the initial reactions in MOF formation is relevant to synthesis strategies. Identical precursors can lead to different MOFs. For example, Goesten[19] et al., calculated that although aluminum NH_2-MIL-53 is thermodynamically favored, aluminum NH_2-MIL-101 is kinetically favored. Multiple competing reactions could occur simultaneously, leading to different intermediates that later combine for nucleation. Finding why in certain conditions relative energy barriers of competing events change to favor certain intermediates over others is key for establishing MOF synthesis strategies.

■ CONCLUSION

A series of reactions that combine chromium ions and terephthalate linkers into the secondary building unit of MIL-101 is proposed. Initial reactions leading to the formation of the metal cluster are followed by linker addition reactions that proceed in different paths due to structural isomers, resulting in SBU formation rates similar to MOF nucleation rates.

The highest energy barrier of the process comes from the initial reactions where chromium atoms assemble with terephthalate linkers to form the metal center. The linkers play a key role as the carboxylate groups join chromium atoms together prior to the formation of bridging oxygens. Linker addition reactions have lower barriers, and these vary with the structural rearrangements each isomer undergoes during the process of removing water molecules to open a binding site in the metal center for linker addition. The lowest energy barrier path identified supports the addition of all three first linkers on the same face of the metal center plane, followed by those in the opposite direction.

Because a single SBU is the building block of many different MOF structures, finding the formation reaction path, mechanisms, and kinetics of an SBU can be used for predicting synthesis kinetics of different MOFs. Also, detailed knowledge of the formation reaction path of an SBU is an initial step toward controlled multiple-step synthesis and deliberate design of new compounds.

With knowledge of the soluble intermediates, new synthesis strategies can be developed, and known metal−linker combinations can lead to new 3D materials, previously kinetically inaccessible. Intermediates, whether $\mathbf{ML_3}$ species or SBUs, will likely assemble into a MOF. Their relative populations are determined by the reaction sequence, and energy barriers that will be affected by solvent and anions present from metal precursors. Different distribution of intermediates could result in defects in the final MOF or different MOF structures. The ratio between intermediates can be influenced by the nature of the solvent, thermodynamically by relative solvation energies, and kinetically by stabilization or steric hindrance at transition states. This can be built into a full-scale micro kinetic model that accounts for solvent as a parameter while utilizing the reaction barriers determined in this work. The effects of solvents and resulting MOF structures can then be predicted based on the reaction series here presented.

Furthermore, different metals and their coordination with ligands and solvent will result in different MOFs. The inert nature of chromium, which causes strong binding with its terephthalate and water ligands, directly affects the energy barriers and reaction series proposed. If a MIL-101 topology is desired with a different metal that is more labile than chromium, the weaker binding could be compensated with a different solvent such that the reaction series toward a MIL-101 topology is favored over other reactions that would occur with weaker metal−ligand interactions.

The ultimate goal is having a thorough and precise knowledge of MOF formation so that synthesis conditions, final structure, and desired properties can be predicted *a priori*. Such a goal is far from being attained due to numerous competing and simultaneous physical and chemical events taking place. This work proposes a synthetic pathway for the formation of an SBU, a key element in rate-limiting MOF nucleation, and helps elucidate how they are constructed. This reaction series and their mechanisms provide a reference to understand the precise role that anions from metal precursors, pH, and solvents can have in MOF synthesis in future studies.

■ ASSOCIATED CONTENT

Supporting Information

Energies, solvation model, electronic spin states, extended reactions, metadynamics, isomer transformation reactions, kinetic model, alternate reaction series. This material is available free of charge via the Internet at http://pubs.acs.org/.

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Notes

The authors declare no competing financial interest.

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■ ABBREVIATIONS

DFT:Density Functional Theory; MIL:Material form Institute Lavoisier; MOF:Metal Organic Framework; NEB:Nudged Elastic Band; SBU:Secondary Building Unit

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